

Core Module Testing Tools

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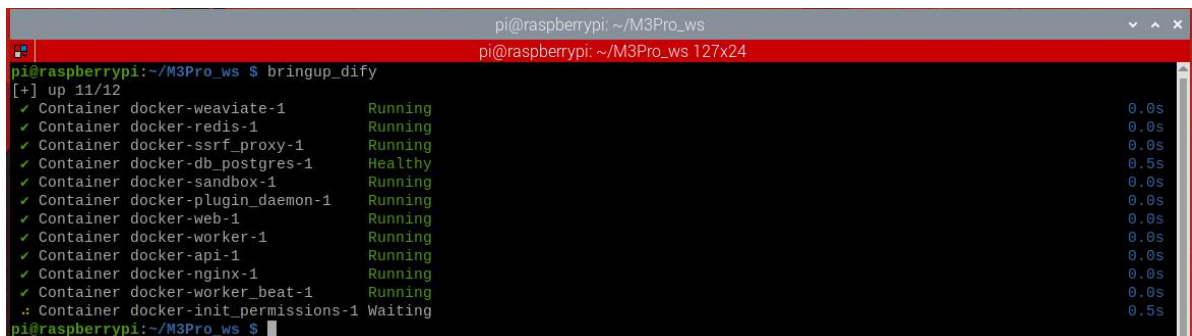
1. Course Content
2. Start the Dify service
3. Enter the test tool path
4. Test if the robot can access Dify normally
5. Online Voice Service for International Users
 - 5.1 iFlytek Speech Recognition Service
 - 5.2 iFlytek Speech Synthesis Service

1. Course Content

- muto-llm provides minimal test programs for testing the core functional modules of a large model, used to quickly locate problems in abnormal situations.

2. Start the Dify service

```
bringup_dify
```

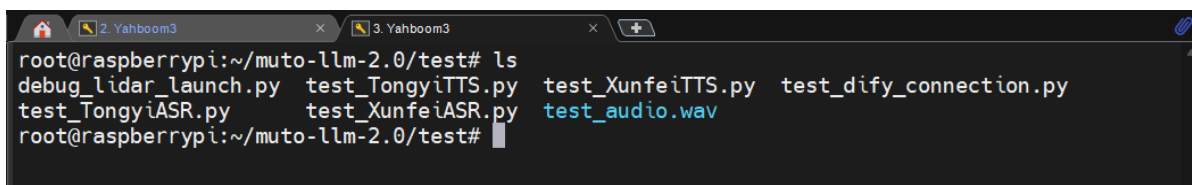


```
pi@raspberrypi: ~/M3Pro_ws
pi@raspberrypi: ~/M3Pro_ws $ bringup_dify
[+] up 11/12
✓ Container docker-weaviate-1      Running      0.0s
✓ Container docker-redis-1        Running      0.0s
✓ Container docker-ssrf_proxy-1    Running      0.0s
✓ Container docker-db_postgres-1   Healthy     0.5s
✓ Container docker-sandbox-1       Running      0.0s
✓ Container docker-plugin_daemon-1 Running      0.0s
✓ Container docker-web-1           Running      0.0s
✓ Container docker-worker-1        Running      0.0s
✓ Container docker-api-1           Running      0.0s
✓ Container docker-nginx-1         Running      0.0s
✓ Container docker-worker_beat-1   Running      0.0s
✗ Container docker-init_permissions-1 Waiting      0.5s
pi@raspberrypi:~/M3Pro_ws $
```

3. Enter the test tool path

```
cd /root/muto-llm-2.0/test
ls
```

- Test tools are as follows:



```
root@raspberrypi:~/muto-llm-2.0/test# ls
debug_lidar_launch.py  test_TongyiTTS.py  test_XunfeiTTS.py  test_dify_connection.py
test_TongyiASR.py      test_XunfeiASR.py  test_audio.wav
```

4. Test if the robot can access Dify normally

- When the program displays that the model service is unavailable, and it is unclear whether the robot's inability to access Dify is due to an incorrect network address, you can test this:

```
python3 test_dify_connection.py
```

If the robot displays the following information normally, it proves that the robot is connected to Dify normally. Otherwise, it may be that Dify is not running or the Dify program has crashed and needs to be restarted.

```
debug_lidar_launch.py test_TongyiTTS.py test_XunfeiTTS.py test_dify_connection.py
test_TongyiASR.py test_XunfeiASR.py test_audio.wav
root@raspberrypi:~/muto-llm-2.0/test# python3 test_dify_connection.py
Testing Dify connection...
Host: http://192.168.12.23
URL: http://192.168.12.23/v1/parameters
Sending GET request...
Status Code: 200
Response Body: {"opening_statement":"","suggested_questions":[],"suggested_questions_after_answer":{"enabled":false},"speech_to_text":{"enabled":false},"text_to_speech":{"enabled":false},"language":"","voice":"","retriever_resource":{"enabled":true},"annotation_reply":{"enabled":false},"more_like_this":{"enabled":false},"user_input_form":[{"file":{"allowed_file_extensions":[],"allowed_file_types":["image"],"allowed_file_upload_methods":["local_file","remote_url"],"default":"","hint":"","label":"image_file","max_length":48,"options":[],"placeholder":"","required":false,"type":"file","variable":"image_file"}},{"text-input":{"default":"TEXT_VERSION","hint":"","label":"request_id","max_length":256,"options":[],"placeholder":"","required":false,"type":"text-input","variable":"request_id"}},{"text-input":{"default":"192.168.12.1","hint":"","label":"ip_address","max_length":48,"options":[],"placeholder":"","required":true,"type":"text-input","variable":"ip_address"}}],"sensitive_word_avoidance":{"enabled":false},"file_upload":{"image":{"enabled":false,"number_limits":3,"transfer_methods":["local_file","remote_url"]},"enabled":false,"allowed_file_types":["image"],"allowed_file_extensions":[".JPG",".JPEG",".PNG",".GIF",".WEBP",".SVG"],"allowed_file_upload_methods":["local_file","remote_url"],"number_limits":3,"fileUploadConfig":{"file_size_limit":15,"batch_count_limit":5,"image_file_size_limit":10,"video_file_size_limit":100,"audio_file_size_limit":50,"workflow_file_upload_limit":10,"image_file_batch_limit":10,"single_chunk_attachment_limit":10}},"system_parameters":{"image_file_size_limit":10,"video_file_size_limit":100,"audio_file_size_limit":50,"file_size_limit":15,"workflow_file_upload_limit":10}}
[✓] Connection successful!
root@raspberrypi:~/muto-llm-2.0/test#
```

5. Online Voice Service for International Users

5.1 iFlytek Speech Recognition Service

- If online speech recognition encounters errors, you can test the Baillian speech recognition service separately.

```
python test_XunfeiASR.py
```

- After running, it will perform speech recognition tests on the preset test audio.

```
root@raspberrypi:~/muto-llm-2.0/test# python3 test_TongyiASR.py
=== Testing Alibaba (Tongyi) ASR ===
VoiceInterface initialized with region: china, language: en, STT provider: alibaba, TTS provider: alibaba
STT Provider alibaba
Speech to text initialized successfully with alibaba provider
Recognizing audio file: /root/muto-llm-2.0/test/test_audio.wav
[User] 我在请吩咐。
Thinking... / [✓] Recognition successful!
Result: 我在请吩咐。
root@raspberrypi:~/muto-llm-2.0/test#
```

5.2 iFlytek Speech Synthesis Service

- If errors occur during online speech synthesis, you can test Baillian's speech recognition service separately.

```
python test_XunfeiTTS.py
```

After running, it will first perform speech synthesis on the default text, and then play the corresponding audio.

```
root@raspberrypi:~/muto-llm-2.0/test# python3 test_TongyiTTS.py
=== Testing Alibaba (Tongyi) TTS ===
VoiceInterface initialized with region: china, language: en, STT provider: alibaba, TTS prov
ider: alibaba
TTS Provider alibaba
aplay verified: aplay
Synthesizing text: 你好，这是阿里语音合成测试。欢迎使用Muto机器人。
[Muto reply] 你好，这是阿里语音合成测试。欢迎使用Muto机器人。
█
```